



SKULLBONEZ

CORE

FIRST-TIME USER MANUAL

Explore. Build a scene. Test what happens.

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Your first session

First-time user manual and keyboard reference

Windows release 2026.09.18

1. Extract, then launch

Extract the entire release ZIP to a folder you can write to. Run **Launch SkullbonezCore.cmd**. Keep the executable, DLLs, and **SkullbonezData** folder together. You do not need Visual Studio, Python, or a source checkout.

2. Find the controls

Use the header's **skull / Tools** button to open the tool drawer. The **Scene** browser loads scenes and creates new ones. Hover over a control for help; scroll inside panels to reach controls below the visible area.

3. Choose how to view the workspace

Canvas gives the scene more room. The **Editor** layout docks editing panels around it. Changing the layout changes the arrangement of panels; the separate **Editor mode** toggle controls whether you are authoring objects.

Start with these controls

Control	What it does
F	Toggle the Inspect/free-camera view.
Hold right mouse + move mouse	Look around the viewport. Release to use the pointer again.
W / A / S / D	Move forward / left / backward / right when camera controls are active.
Shift + movement	Move faster (3x camera travel speed).
Tab	Cycle camera modes.
Backtick (`)	Toggle Editor mode. This is the US-keyboard key below Esc.
0 (zero)	Toggle UI visibility.
P	Enter paused prediction inspection; press again to return to live.

Good to know

For your first build, follow pages 2 and 3. Use the visible Pause/Resume and One tick controls rather than guessing which mode owns a keyboard shortcut.



Build your first scene

Example project: a box and a ball on flat terrain. No JSON editing is required.

1. Open the Scene browser

Click **Scene** in the header, or open **Tools > Scene**. Open the scene dropdown so its text filter is active.

2. Give the scene a new name

Type **my_first_scene**. Use a name that does not already match a scene. The same field filters existing scenes and supplies the name for a new one.

3. Create the draft

Choose **Create new scene**. If there are no matching scenes, pressing **Enter** also creates the typed name. An existing match can load that scene instead, so check the list first. The starter has flat terrain, hidden water, and ordinary downward gravity.

4. Enable editing

Open the **Editor** controls and turn **Editor mode** on, or press **backtick (`)**. Entering Editor mode starts in **Place mode**. Live scene physics is held while you edit.

5. Place a box

Choose **Box** in the **Object** list. Turn **Static object** on for a fixed obstacle, or off for a body that can fall. Move the pointer onto terrain until the preview appears, then click and release the left mouse button to place it.

6. Add a ball

Choose **Ball**, turn **Static object** off, and move the preview near the box. Roll the mouse wheel before placing to raise it above the terrain. Click and release to place it. See page 3 for sizing and rotation.

7. Save the named scene

Return to **Tools > Scene** and click **Save Defaults**. For this editable scene, that writes the scene and its objects to **SkullbonezData/scenes/my_first_scene.scene.json**. A conflicting name receives a numbered suffix.

8. Run it and reopen it

Turn Editor mode off with backtick. If the scene remains paused, use **Resume**. Later, open the Scene browser, filter for **my_first_scene**, and select it. Save a deliberate starting arrangement before running an experiment.

Good to know

Create makes an in-memory draft. Use **Save Defaults** before changing scenes or closing the app. **F2** creates a separate numbered snapshot; it is not the named-scene save action.



Place, select, transform

The pointer acts on the viewport. A click over a panel operates that panel instead.

Mode	Use it for
Place mode ON	Preview and insert a new object. Pick its type from the Object list or the quick palette.
Place mode OFF / Gizmo	Select an existing object, then move, rotate, or scale it. A gizmo is the visible set of transform handles.

Placing objects

Gesture or control	Result
Left click and release	Place the current preview. Placement commits on release.
Hold left mouse + drag	Resize the preview before placing. Boxes use horizontal/vertical drag for footprint dimensions; wheel adjusts height during the drag. Round objects scale uniformly.
Mouse wheel before clicking	Raise/lower the preview in object-sized steps above its terrain placement.
Ctrl + mouse wheel	Rotate the placement preview around the vertical axis in 15-degree steps.
Static object	On: fixed body. Off: dynamic body that can respond to physics.
Terrain align	Align placement to the terrain slope.

Editing existing objects

Turn **Place mode off** (or tap **Alt** while in Editor mode), then left-click an object. Keep **Modify velocity off** when editing its shape or position.

Gesture	Result
Left-drag an axis handle	Move along that axis.
Left-drag a rotation ring	Rotate around that axis.
Ctrl + left-drag an axis handle	Scale along that axis.
Delete	Delete the selected editable object.
Ctrl + Z / Ctrl + Y	Undo / redo. Ctrl + Shift + Z also redoes.

Camera: with Terrain brush off, hold right mouse over the viewport to look around; use WASD to move and Shift for faster travel. Release right mouse to edit again.

Good to know

Undo history is cleared when you leave Editor mode. Save your starting scene and use undo before turning editing off.



Physics, prediction & replay

Start simple: save a scene, run it, inspect what happens, then reload your saved setup.

Control the simulation

Turn **Editor mode off** to let live physics run. Use **Pause / Resume** to hold or release the scene and **One tick** (or **Single step** in Scene controls) to advance a paused simulation once. **Simulation speed** changes the rate of advancement.

The Physics controls provide body properties, simulation settings, and diagnostic overlays. Select an object for body-specific controls. A **static** object stays fixed; a **dynamic** object can move. A sleeping dynamic body may be at rest until something wakes it.

Inspect a predicted future

1. Enter prediction

Press **P** to pause live physics and enable prediction inspection. The interface is shown in a compact state so you can select in the world.

2. Select a body

Left-click a body in the viewport. Allow the prediction to finish before using controls that depend on its result. Prediction paths describe a computed future from the selected state.

3. Read, then return

Use the Replay and Causes controls to inspect available recorded or predicted information. Press **P** again to clear the inspection and return to live simulation. **F7** toggles the Split Future look.

Understand the different saves

Action	What you keep
Scene: Save Defaults	For a new editable scene, saves the named scene, its objects, and current scene state under SkullbonezData/scenes/.
F2	A separate scene snapshot: Scenes/snapshot_*.scene.json. Useful as an extra checkpoint.
F3	A viewport screenshot: Screenshots/screenshot_*.bmp.
Replay: Save / Load recording	Recorded simulation data. A recording is different from an editable scene.

Good to know

Reset rebuilds the scene while preserving live runtime controls. **Reset Defaults** discards live scene edits and reloads authored defaults. Save anything you want to keep before reloading defaults.

Explore the supplied comparisons

Open **Tools > Scene > Solver Lab** and choose a bundled comparison. In Solver Lab, **Space** plays/pauses and the **left/right arrow keys** step backward/forward. These are context-specific controls.



Keyboard & mouse

Shortcuts are context-sensitive. Finish typing in a filter or close a popup before using world controls.

Key / gesture	Action and context
F / Tab	Toggle Inspect camera / cycle camera modes.
Right mouse + mouse movement	Viewport camera look; Terrain brush changes right mouse to lowering terrain.
W A S D / Shift	Camera movement / faster camera travel when camera input is active.
Backtick (`)	Enter or leave Editor mode.
Alt	In Editor mode: switch Place/Gizmo. In prediction, Alt also participates in velocity editing.
Ctrl + Z	Undo an editor edit.
Ctrl + Y or Ctrl + Shift + Z	Redo an editor edit.
Delete	Delete the selected editable object.
Ctrl + wheel	Rotate a placement preview by 15 degrees per wheel step.
Ctrl + left-drag axis	Scale a selected object in Gizmo mode.
P	Toggle paused prediction inspection and return to live.
F7	Toggle Split Future look.
0 (zero)	Show/hide the UI.
F2 / F3	Save a scene snapshot / save a screenshot.
R	Reset the scene. Save changes you want to preserve first.
N / M	Toggle launcher / cycle fire mode while launcher is active.
F1 / Enter	Attached camera only: cycle submode / toggle pin.
Space / left-right arrows	Solver Lab only: play/pause / step backward-forward.
F5 / F6	Toggle the performance histogram / memory overlay.
V / C	Toggle collision visualization / cycle physics debug overlays.
G / O	Toggle broadphase bounds / terrain contact probe.
[/] / 6	Previous/next physics pipeline stage / transparent physics-debug display.
1 / 2 / 3	Freeze animated water / cycle reflection mode / toggle flat-water display.
4 / 5	Show/hide terrain / water.

These are the main first-session controls, not every authoring or developer shortcut. Hover help and the active mode remain the best guide for context-specific tools.



When something feels stuck

Most first-session surprises come from an active mode, a paused scene, or a pointer over a panel.

What you see	What to check
Objects will not move	Turn Editor mode off. Check Pause/Resume. Make sure the body is dynamic rather than Static. P may have left you in prediction inspection.
Clicks keep adding objects	Place mode is on. Switch to Gizmo with the Place mode toggle or Alt before selecting an existing object.
The cursor disappears	Release right mouse. In Place mode, the preview can replace the normal cursor over the world.
No placement preview	Move over terrain inside the viewport, away from tool panels. Check Editor mode and Place mode.
Shortcuts do nothing	Click back into the viewport or finish editing the text field. Close open menus. Check that you are in the correct workspace.
Controls are missing	Scroll inside the panel, unfold its dock, or enlarge the window. Press 0 if the UI is hidden; use the skull/Tools button for the drawer.
My new scene is missing after restart	Creation alone does not save it. Use Scene > Save Defaults and check for its .scene.json file under SkullbonezData/scenes/.
Files do not save or assets are missing	Extract the whole ZIP to a writable folder and use Launch SkullbonezCore.cmd. Do not run the executable from inside the ZIP.

Terrain editing

With Editor mode on, enable **Terrain brush**. Hold **left mouse** to raise terrain or **right mouse** to lower it; roll the **mouse wheel** to adjust brush radius. Turn Terrain brush off before using right mouse for camera look. Save the editable scene after terrain changes.

A useful first practice session

Create **my_first_scene**, place one static box and one dynamic ball, save, leave Editor mode, and watch. Reload your saved defaults, move the ball higher, save another checkpoint with F2, and use P to inspect the resulting prediction.

About this guide

Prepared for **SkullbonezCore 2026.09.18**, main commit **ff65964**. Controls and workflow were checked against the release-era input bindings, editor interaction code, scene creation/save code, and in-app labels. This is a source-checked guide; it focuses on the controls and workflows a new user needs first.

Release and downloads: [SkullbonezCore 2026.09.18](#)